

TCS3351 Cryptography and Data Security
Tutorial 9
Modern Asymmetric Ciphers

1. Given $a = [15 \ 25 \ 50 \ 100 \ 200 \ 400 \ 800]$ and $s = 450$. Find tuple x using inv-knapsackSum algorithm.

Answer:

i	a_i	s	$s \geq a_i$	x_i	$s = s - a_i$
7	800	450	No	0	450
6	400	450	Yes	1	50
5	200	50	No	0	50
4	100	50	No	0	50
3	50	50	Yes	1	0
2	25	0	No	0	0
1	15	0	No	0	0

Hence, the tuple $x = [0,0,1,0,0,1,0]$.

2. Given the super increasing tuple $b = [7 \ 11 \ 23 \ 43 \ 87 \ 173 \ 357]$, $r = 41$, and modulus $n = 1001$, encrypt and decrypt the letter 'a' using the knapsack cryptosystem. Use $[7 \ 6 \ 5 \ 1 \ 2 \ 3 \ 4]$ as the permutation table.

Answer:

$$\text{ACSII}(a) = (1100001)_2 = [1,1,0,0,0,0,1]$$

Encryption:

$$7*41 = 287, 11*41 = 451 \dots$$

$$\text{Step 1: } t = b \times 41 \pmod{1001} = [287 \ 451 \ 943 \ 762 \ 564 \ 86 \ 623].$$

$$\text{Step 2: Permutation } [7 \ 6 \ 5 \ 1 \ 2 \ 3 \ 4] = [623 \ 86 \ 564 \ 287 \ 451 \ 946 \ 762]$$

No	1	2	3	4	5	6	7	
	287	451	943	762	564	86	623	
Permutation	7	6	5	1	2	3	4	
	623	86	564	287	451	943	762	
x_i	1	1	0	0	0	0	1	

Hence, the ciphertext, $s = 623 + 86 + 762 = \mathbf{1471}$.

Decryption:

$$\text{Step 1: Inverse } r^{-1} \pmod{n} = 41^{-1} \pmod{1001} = 293$$

q	a	b	r	t_1	t_2	t
24	1001	41	17	0	1	-24

2	41	17	7	1	-24	49
2	17	7	3	-24	49	-122
2	7	3	1	49	-122	293
3	3	1	0			

Step 2: $s' = s \times r^{-1}(\text{mod } n) = 1471(293) \equiv 573 (\text{mod } 1001)$.

Step 3: $x' = [0,0,0,1,0,1,1]$

i	a_i	s	$s \geq a_i$	x_i	$s = s - a_i$
7	357	573	Yes	1	216
6	173	216	Yes	1	43
5	87	43	No	0	43
4	43	43	Yes	1	0
3	23	0	No	0	0
2	11	0	No	0	0
1	7	0	No	0	0

Step 4: Permutation [7 6 5 1 2 3 4], $\text{Per}(x') = [1,1,0,0,0,0,1]$

Hence, the plaintext $[1,1,0,0,0,0,1] = (1100001)_2 = a$.

3. Perform key generation, encryption and decryption using RSA algorithm for the following problem.

$$p = 11; q = 13; e = 7; \text{Message} = FCI$$

Answer:

Key Generation:

- $n = pq = (11)(13) = 143$.
- $\phi(n) = (p-1)(q-1) = (10)(12) = 120$.
- $d \equiv e^{-1} (\text{mod } \phi(n)) \equiv 7^{-1} (\text{mod } 120) \equiv -17 (\text{mod } 120) \equiv 103 (\text{mod } 120)$.

q	a	b	r	t_1	t_2	t
17	120	7	1	0	1	-17
7	7	1	0			

- d) Public key: (7,143), Private key: (103).

Encryption: Let plaintext be $P(F, C, I) = [5, 2, 8]$. Ciphertext, $C \equiv P^7 (\text{mod } 143)$. Hence, the ciphertext $C = [47, 128, 57]$.

P	$C \equiv P^7 (\text{mod } 143)$
5	47
2	128

8	57
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Decryption: Plaintext, $P \equiv C^{103} \pmod{143}$. Hence, the plaintext $P = [5, 2, 8] = [F, C, I]$.

P	$P \equiv C^{103} \pmod{143}$
5	$47^{103} \pmod{143}$
2	128
8	57

Fast exponentiation algo.

103 $\Rightarrow$ 110 0111

1st. Change the exponent 103 to binary: $103 = 64 + 32 + 4 + 2 + 1 = 110\ 0111$

2nd. Label all variable $a = \text{base} = 47$, $n = 143$. $Y = 1$ (initially)

Draw the table. Do the calculation.

Bit	Multiply if $x_i = 1$, $Y = (a * y) \pmod{n}$	Square $A = a * a \pmod{n}$
1 lsb	$Y = 47 * 1 = 47$	$a = 47.47 \pmod{143} = 64$
1	$Y = 64 * 47 \pmod{143} = 5$	$a = 64.64 \pmod{143} = 92$
1	$Y = 5 * 92 \pmod{143} = 31$	$a = 92.92 \pmod{143} = 27$
0	31	$a = 27.27 \pmod{143} = 14$
0	31	$a = 14.14 \pmod{143} = 53$
1	$Y = 31 * 53 \pmod{143} = 70$	$a = 53.53 \pmod{143} = 92$
1 msb	$Y = 92 * 70 \pmod{143} = 5$	$a = 92.92 \pmod{143} = 27$

4. Perform key generation, encryption and decryption using RSA algorithm for the following problem.

$$p = 7; q = 11; e = 17; M = 8$$

Answer:

Key Generation:

- $n = pq = (7)(11) = 77$.
- $\phi(n) = (p - 1)(q - 1) = (6)(10) = 60$.
- $d \equiv e^{-1} \pmod{60} \equiv 17^{-1} \pmod{60} \equiv -7 \pmod{60} \equiv 53 \pmod{60}$.

q	a	b	r	t_1	t_2	t
3	60	17	9	0	1	-3
1	17	9	8	1	-3	4

1	9	8	1	-3	4	-7
8	8	1	0			

d) Public key: (17,77), Private key: (53).

Encryption: Ciphertext $C \equiv P^{17} \pmod{77} \equiv 8^{17} \pmod{77} \equiv 57$.

Decryption: Plaintext $P \equiv C^{53} \pmod{77} \equiv 57^{53} \pmod{77} \equiv 8$.

$$(8^{17})^{53} \pmod{77} = 8^{17 \cdot 53 \bmod 60} \pmod{77} = 8.$$

Because $17 \cdot 53 \bmod 60 = 1$. The inverse of 17 is 53.

$$d \equiv e^{-1} \pmod{\phi(n)}$$

$$Ed = 1 \bmod \phi(n)$$

References:

1. Forouzan, B. A. & Mukhopadhyay, D. (2015). Cryptography & Network Security (3rd ed.). McGraw-Hill.
2. William Stallings, "Cryptography and Network Security – Principles and Practice", Seventh Edition, Prentice Hall, 2017